AerOptima

Predicting Flight Delays

An End-to-End Data Project

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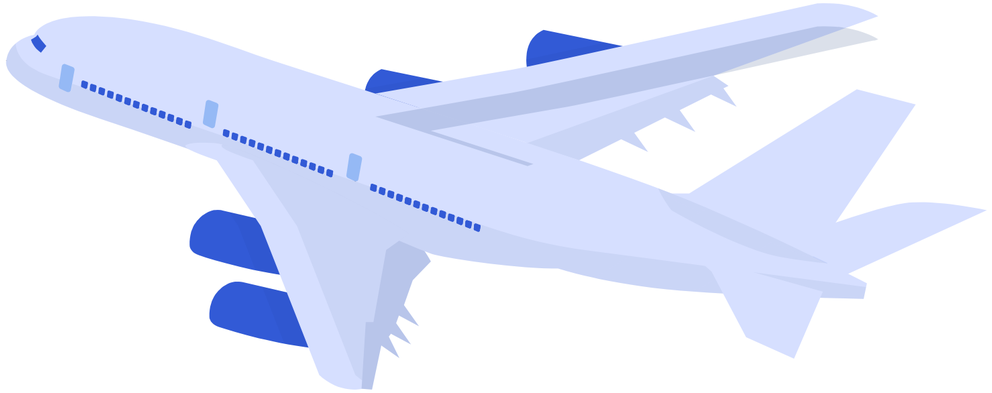

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The Challenge & The Data

Predicting the Unpredictable

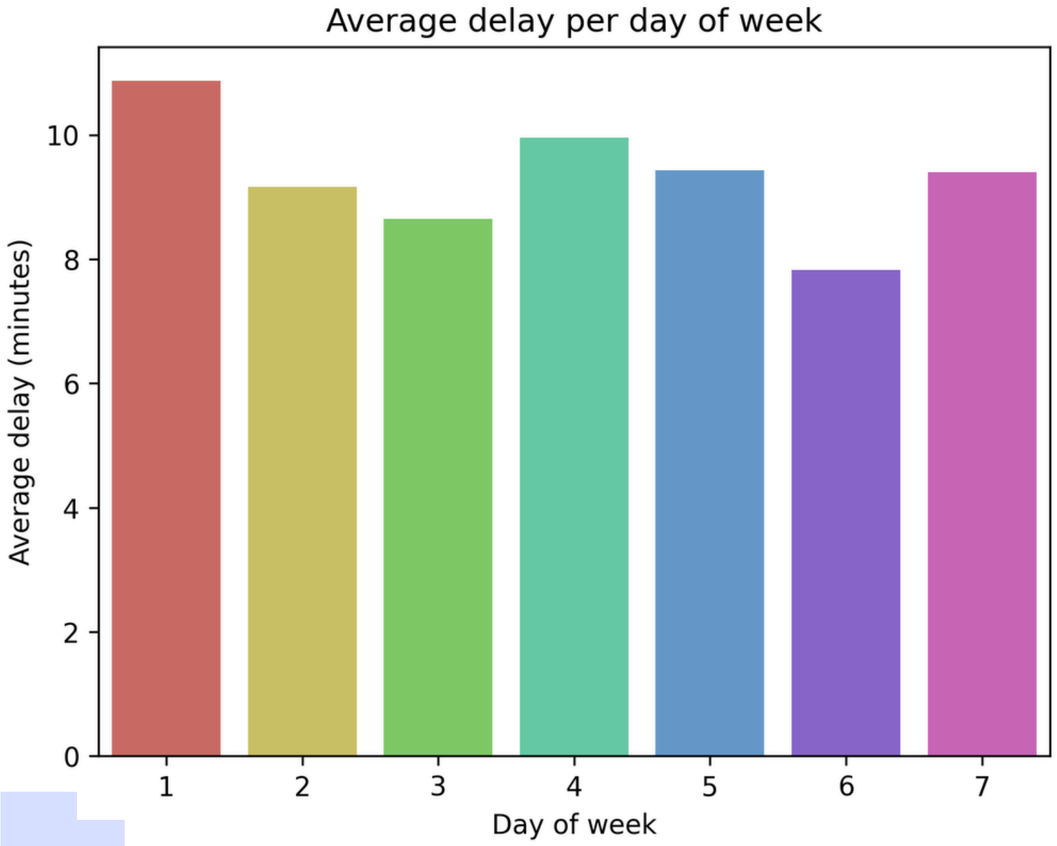
Objective: Predict a flight's arrival delay using only information available before takeoff

The Dataset: 2015 Flight Delays and Cancellations (U.S. Department of Transportation)

Scope: Millions of domestic US flights containing schedules, airlines, distances, and delay causes

Exploratory Data Analysis (EDA)

Finding the Signal in the Skies



- Analyzed temporal patterns and airline performance

- **Key Insight:** Delays follow structural patterns:
  - ➡ For example, the day of the week significantly impacts the average delay, showing air traffic congestion trends

The Machine Learning Trap

The "Too Good To Be True" Model

Initial Result: Our first XGBoost model achieved an incredibly low Mean Absolute Error (MAE) of **3.5 minutes**

The Problem: Data Leakage

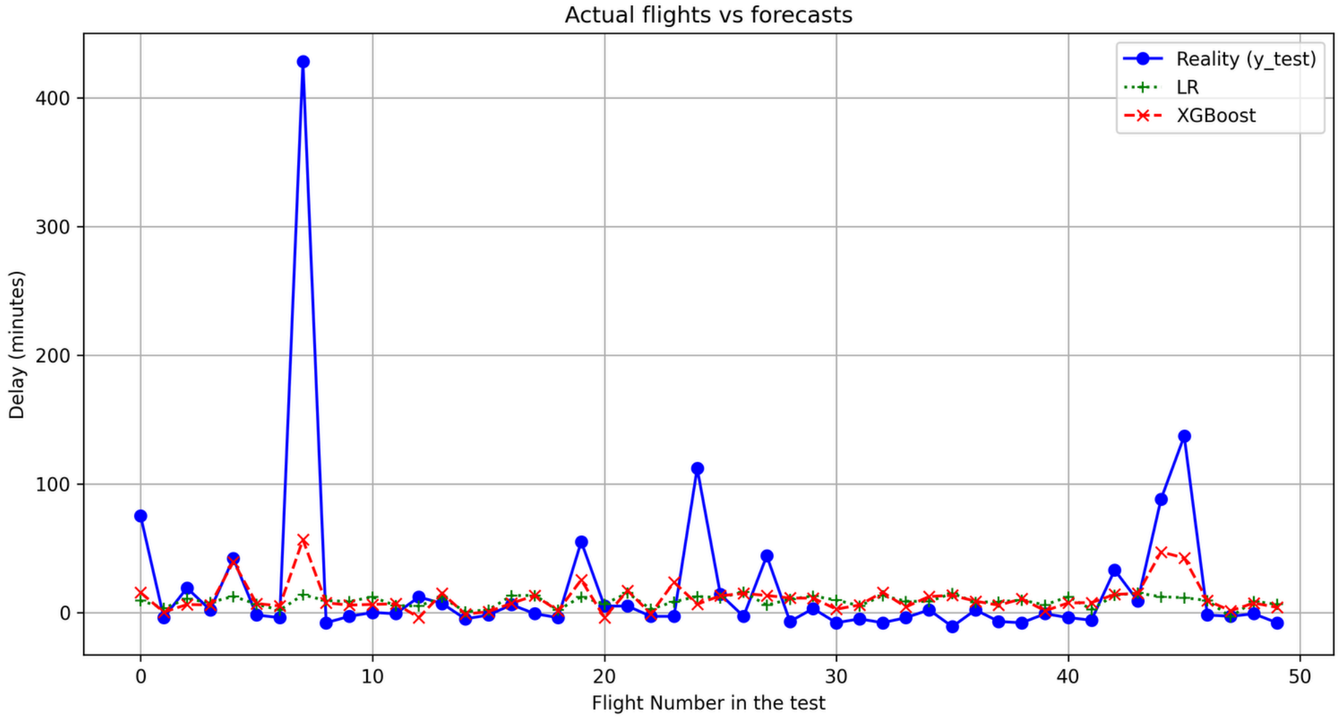
The Investigation: Feature Importance revealed the model was "cheating" by using variables created after the flight landed (e.g., *LATE_AIRCRAFT_DE*LAY, *WHEELS_ON*)

Building an Honest Model

Real Predictions with XGBoost

- Cleaned the dataset by dropping all "future" variables

- **Final Result:** A realistic MAE of **~17 minutes**



- **Insight:** The model successfully predicts the baseline structural delays (the signal) but wisely ignores unpredictable, exceptional events (the noise)

Conclusion & Next Steps

Takeaways and Future Work

Tree-based models (**XGBoost**) easily outperform rigid models (**Linear Regression**) for this complex task

Identifying **Data Leakage** is the most critical step for real-world deployment

Future Work: Switch to a Classification approach (On Time vs. Delayed) or enrich the dataset with historical weather forecasts